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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**

Off-Campus Centre of Nitte(Deemed to be University)

**Fifth Semester Minor Degree Examinations**

December 2025

**CS2006-1- FUNDAMENTALS OF DATA STRUCTURES**

(For Minor Degree)

Duration: 3 Hours

Max. Marks:100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. Which of the following header files must necessarily be included to use dynamic memory allocation functions?  
A) memory.h  
B) stdio.h  
C) stdlib.h  
D) dos.h
2. Which of the following return-type cannot be used for a function in C?  
A) char \*  
B) struct  
C) void  
D) None of the options
3. What is the correct way to assign one structure to another?  
struct Point { int x, y; };  
struct Point p1 = {1, 2}, p2;  
A) p2 = p1;  
B) p2 == p1;  
C) p2.equals(p1);  
D) Structures cannot be assigned directly
4. Why is it often preferable to pass structures by reference instead of by value in C?  
A) Passing by value is faster because it avoids memory allocation.  
B) Passing by reference avoids making a copy of the structure, which can be costly if the structure is large.  
C) Passing by value is required for structures that contain arrays.  
D) Passing by reference ensures that the structure members are always properly aligned.
5. Consider the following stack operations: push(5), push(10), push(20), pop(), push(15), pop(). What will be the final state of the stack?  
A) [5, 10]  
B) [5, 10, 15]  
C) [5, 15]  
D) [10, 15]
6. Which of the following is not application of stack  
A) Recursion  
B) Reversing a string  
C) Evaluation of postfix expression  
D) Accessing element in the middle of the list
7. Disadvantage of stacks  
A) Allows elements to be inserted through one end top  
B) Allows elements to be deleted through one end top  
C) Display works in the order from the top end  
D) Stack does not allow random access to element
8. Which of the following problems can't be solved using recursion?  
A) Factorial of a number  
B) Nth fibonacci number  
C) Length of a string  
D) Problems without base case
9. Principle followed by queue is  
A) LIFO  
B) FIFO  
C) LOFI  
D) Random
10. A circular queue is implemented using an array of size 10. The array index starts with 0, front is 6, and rear is 9. The insertion of next element takes place at the array index.....  
A) 0  
B) 7  
C) 9  
D) 10

11. Which data structure is commonly used to implement the event-driven simulation of complex systems, such as in computer network simulations or traffic simulations?  
 A) Stack B) Tree  
 C) Array D) Queue
12. Which of the following conditions checks the underflow condition in the queue?  
 A) front = -1 B) rear = MAX  
 C) rear = -1 D) rear = front
13. Which of the following is true about the head node in a singly linked list?  
 A) It contains the largest value in the list B) It points to the first node of the list  
 C) It points to the last node of the list D) It always Null
14. How do you find the length of a singly linked list?  
 A) By traversing the list and counting the nodes B) By accessing the length property  
 C) By using a counter variable during insertion and deletion D) Both a and c
15. In which scenario is the Linked List an excellent data structure?  
 A) When you need to insert elements at the end of the B) When you need to insert elements in the middle of the list  
 C) When you need to access elements in sequential order D) When you need quick access to elements by index
16. Given pointer to a node X in a singly linked list. Only one pointer is given, pointer to head node is not given, can we delete the node X from given linked list?  
 A) Possible if X is not last node B) Possible if size of linked list is even  
 C) Possible if size of linked list is odd D) Possible if X is not first node
17. Which of the following is non-linear data structure?  
 A) tacks B) List  
 C) Strings D) Trees
18. The number of edges from the root to the node is called \_\_\_\_\_ of the tree.  
 A) Height B) Depth  
 C) Length D) Width
19. What is a complete binary tree?  
 A) Each node has exactly zero or two children B) A binary tree, which is completely filled, with the possible exception of the bottom level, which is filled from right to left  
 C) A binary tree, which is completely filled, with the possible exception of the bottom level, which is filled from left to right D) A tree in which all nodes have degree 2
20. In a binary tree, the in-order traversal of nodes results in the sequence 4, 2, 5, 1, 3. What is the root of the binary tree?  
 A) 1 B) 2  
 C) 3 D) 4

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                                  | Marks | BT* | CO* | PO* |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a)    | Discuss tower of hanoi with four disks and write its implementation.                                                                                                                                             | 6     | L*2 | 2   | 1   |
| b)       | Define stack. List any four applications of stack. Build a C program to check a string is palindrome or not using stack.                                                                                         | 10    | L3  | 2   | 1   |
| 2. a)    | Discuss any four advantages of pointers. Build a C program to count the number of vowels and consonants in a string using pointers.                                                                              | 8     | L3  | 1   |     |
| b)       | Write an algorithm to convert a parenthesized infix expression to postfix expression. Apply the algorithm and show the contents of a stack during conversion for the expression:<br>((A + (B - C) * D) \$ E + F) | 8     | L3  | 2   | 1   |
| 3. a)    | Describe nested structures with suitable example                                                                                                                                                                 | 6     | L2  | 1   | 1   |

- b) Build a c program using structures for reading details of N number of employees. Use nesting structures to store the address of an employee. (The data fields of an employee structures is as follows: employee number, employee name, address, salary.

Address contains: house number, street, place, pin code)

- |                                            |                                                                                                     |   |    |   |   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------|---|----|---|---|
| i.                                         | Calculate the average salary, and display details of employees whose salary is above average salary | 6 | L3 | 1 | 1 |
| c) Solve the following postfix expressions |                                                                                                     |   |    |   |   |
| i.                                         | 6 5 2 3 + 8 * + 3 + *                                                                               |   |    |   |   |
| ii.                                        | 100 200 + 2 % 5 * 7 +                                                                               | 4 | L3 | 2 | 1 |

#### Unit – II

- |       |                                                                                                                       |    |    |   |   |
|-------|-----------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 4. a) | Describe priority queue with suitable example. Build a C program to design a queue. Provide the following operations: |    |    |   |   |
| i.    | Insert                                                                                                                |    |    |   |   |
| ii.   | Delete                                                                                                                |    |    |   |   |
| iii.  | Display                                                                                                               | 10 | L3 | 3 | 1 |
| b)    | Discuss the advantages of linked list.                                                                                | 4  | L2 | 4 | 1 |
| c)    | Write a short note on Self referential Structures.                                                                    | 2  | L2 | 4 | 1 |

5. a) Design/ Build a C functions for the following related to Singly Linked List(SLL)

i. Searching an element in the list (3M)

ii. Merge of two singly linked list (5M)

Example:

L1: 10→20→30→40→NULL

L2: 1→2→3→4→NULL

After merging

10→1→20→2→30→3→40→4→NULL

- |       |                                                                                                                                                                                                               |   |    |   |   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| b)    | Define queue. State the limitations of ordinary queue. Explain how do you overcome the limitations of ordinary queue by specifying the required C-statements and diagrammatic representation using an example | 8 | L3 | 4 | 1 |
| 6. a) | Build a C functions for the following related to Singly Linked List (SLL)                                                                                                                                     |   |    |   |   |
| i.    | delete_front()                                                                                                                                                                                                |   |    |   |   |
| ii.   | insert_rear()                                                                                                                                                                                                 | 8 | L2 | 4 | 1 |
| b)    | Write a C functions for the following related to Circular linked list with header node                                                                                                                        |   |    |   |   |
| i.    | Insert node at front end                                                                                                                                                                                      |   |    |   |   |
| ii.   | Delete a node from front end                                                                                                                                                                                  | 8 | L2 | 4 | 1 |

#### Unit – III

- |       |                                                                                                          |   |    |   |   |
|-------|----------------------------------------------------------------------------------------------------------|---|----|---|---|
| 7. a) | Write a C program to implement a binary tree of integers and perform the following traversal techniques. |   |    |   |   |
| i.    | In-order traversal                                                                                       |   |    |   |   |
| ii.   | Post-order traversal                                                                                     |   |    |   |   |
| iii.  | Pre-order traversal.                                                                                     | 8 | L3 | 5 | 1 |
| b)    | Discuss array and linked list representation of Binary Trees.                                            | 8 | L2 | 5 | 1 |
| 8. a) | Define Binary Trees. List and explain any 4 types of binary trees                                        | 8 | L2 | 5 | 1 |
| b)    | Write a C function to                                                                                    |   |    |   |   |
| i.    | Find maximum value in a Binary search tree (BST)                                                         |   |    |   |   |
| ii.   | To count the number of nodes in BST                                                                      |   |    |   |   |
| iii.  | To count the number of leaf nodes in BST                                                                 | 8 | L3 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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